

REGIONAL MARITIME UNIVERSITY
FACULTY OF ENGINEERING AND APPLIED SCIENCES
BIT27B, BME27B, BNS27B, BEE27B, BCS27B, BCE27B, BMT27B
END OF FIRST SEMESTER EXAMINATIONS
BSMA 101: MATHEMATICS I **Duration: 2 hours 30 minutes**

Answer **THREE (3)** Questions only. **Question 1 is compulsory.**

1. (a) Find the first derivative of the following functions:

(i) $y = (\tan x)^x$ [3 marks]

(ii) $y = \frac{1 + \sin^2 x}{1 + \cos^2 x}$ [3 marks]

(iii) $x^2 + xy + y^2 = a^2$ where a is a constant find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. [4 marks]

(b) Find the equation of the normal to the curve $y = 2x^3 + 3x^2 - 2x - 3$ at the point $(1, 0)$. [3 marks]

(c) The height, h in metres, of an object after t seconds after being fired into the air is given by $h = 19.2t - 4.8t^2$.

Find: (i) the initial velocity of the object. (i.e at time $t = 0$)

(ii) the height it attained when its velocity is one half of its initial velocity.

[NOTE: velocity, $V = \frac{dh}{dt}$] [4 marks]

(d) Express $Z = 4 + 3i$ in polar form. [2 marks]

(e) Simplify $\frac{2-i}{1-i}$ [2 marks]

(f) Evaluate $|a \times b|$ if $a = i + 2j + 2k$ and $b = 2i - j + 2k$. [3 marks]

(g) Sketch the graph of the rational function $y = \frac{3x-1}{x+4}$ [3 marks]

(f) If $y = u^2 + 1$, $u = \sqrt{s}$, $s = x^3$, find $\frac{dy}{dx}$ at $x = 2$. [3 marks]

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2. (a) Find the coordinates of the turning points of the curve $y = \frac{x^3}{3} + \frac{x^2}{2} - 6x + 9$. Hence find the maximum and minimum values of y . [4 marks]
- (b) Resolve $\frac{x-1}{(x+1)(x^2+1)}$ into partial fraction. [5 marks]
- (c) Find the odd and even parts of the function, $f(x) = \frac{1}{1+x}$. [4 marks]
- (d) Calculate the work done by a force 15 N acting on a particle in the direction of the vector $\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$ which moves a particle from $P(1, 1, 3)$ to $Q(4, 5, 1)$. [2 marks]
3. (a) $f(x) = 7x - 2$ $g(x) = x^2 + 1$ $h(x) = 3^x$
- (i) Find $gh(2)$ [2 marks]
- (ii) If $gg(x) = ax^4 + bx^2 + c$, find a , b and c . [3 marks]
- (iii) Find x when $hf(x) = 81$ [3 marks]
- (iv) Find $f^{-1}(h)$ [2 marks]
- (b) If $4 + 2i = r(\cos\theta + i \sin\theta)$ where r is a positive real number and θ is an acute angle, find r and θ . [5 marks]
4. (a) Define the scalar product $\mathbf{a} \cdot \mathbf{b}$ of two vectors $\mathbf{a}$ and $\mathbf{b}$ having acute angle θ between them.
- (i) show that the vectors $\mathbf{u} = 2\mathbf{i} + 3\mathbf{j} - 6\mathbf{k}$, $\mathbf{v} = 6\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ and $\mathbf{w} = 3\mathbf{i} - 6\mathbf{j} - 2\mathbf{k}$ are mutually perpendicular.
- (ii) Find the unit vectors $\hat{\mathbf{u}}$, $\hat{\mathbf{v}}$ and $\hat{\mathbf{w}}$. [6 marks]
- (b) If $y = \sin t$ and $x = \cos 2t$ find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. [4 marks]
- (c) A sheet of metal measures 240 mm by 90 mm . Four equal squares each of side $x \text{ mm}$ are cut from the sheet, one from each corner. The sides are then turned up to make an open box. Find the size of the squares to make the box with the greatest volume. [5 marks]

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5. (a) An inverted right circular cone of vertical angle 60° is collecting water from a tap at a steady rate of $18\pi \text{ cm}^3/\text{s}$. Find the rate at which the depth (h) of water is rising. [4 marks]

(b) Evaluate the following limits;

(i) $\lim_{x \rightarrow -1} \left(\frac{(x+1)e^{3x+2}}{x^2-1} \right)$

(ii) $\lim_{n \rightarrow \infty} \left\{ \frac{n^3-2}{2n^3+3n-4} \right\}$

(iii) $\lim_{x \rightarrow 0} \left(\frac{x \cos x - \sin x}{x^3} \right)$

(iv) $\lim_{x \rightarrow \frac{\pi}{2}} (2x^2 \cos x [3x - \frac{\pi}{2}])$

(v) $\lim_{x \rightarrow 1} \left(\frac{x^2-1}{x-1} \right)$

[7 marks]

- (c) If $\frac{7+i}{3+4i} = \frac{a+bi}{4-i}$, where a and b are real, find the values of a and b. [4 marks]

B. A.

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